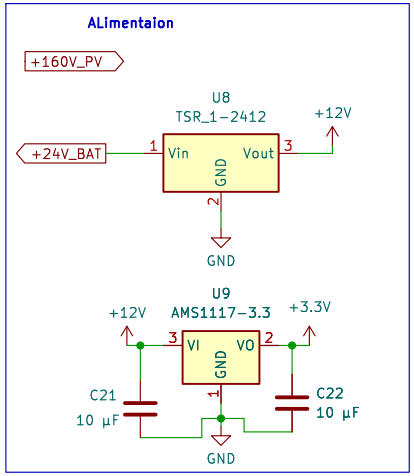
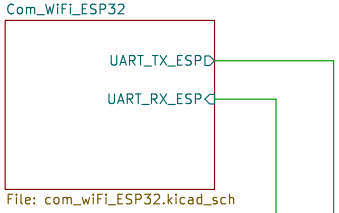
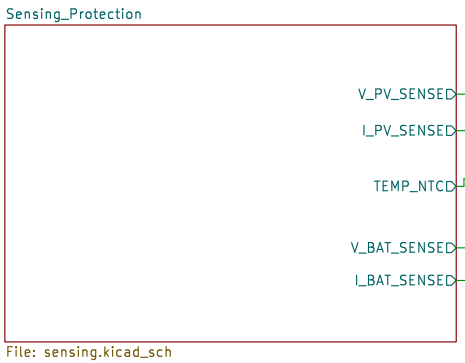


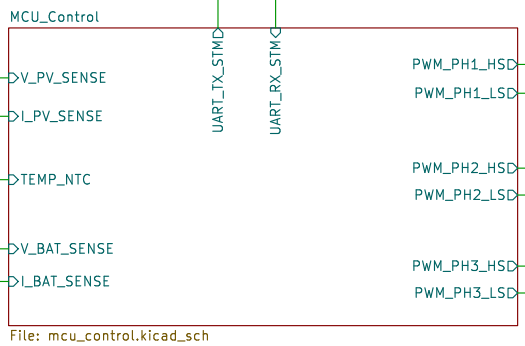


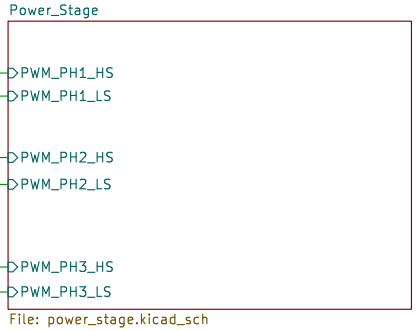

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LEXIQUE & COMPOSANTS CLÉS DU PROJET

--- 1. ABRÉVIATIONS ESSENTIELLES ---
MPPT : Maximum Power Point Tracking (Algo d'optimisation solaire)
PWM  : Pulse Width Modulation (Signal de commande du hacheur)
ADC  : Analog to Digital Converter (Lecture analogique des sondes)
HS/LS: High-Side (Haut) / Low-Side (Bas) désignant les MOSFETs
ESR  : Equivalent Series Resistance (Résistance interne d'un condensateur)
NTC  : Negative Temperature Coefficient (Sonde thermique de sécurité)
SWD  : Serial Wire Debug (Port de programmation du microcontrôleur)

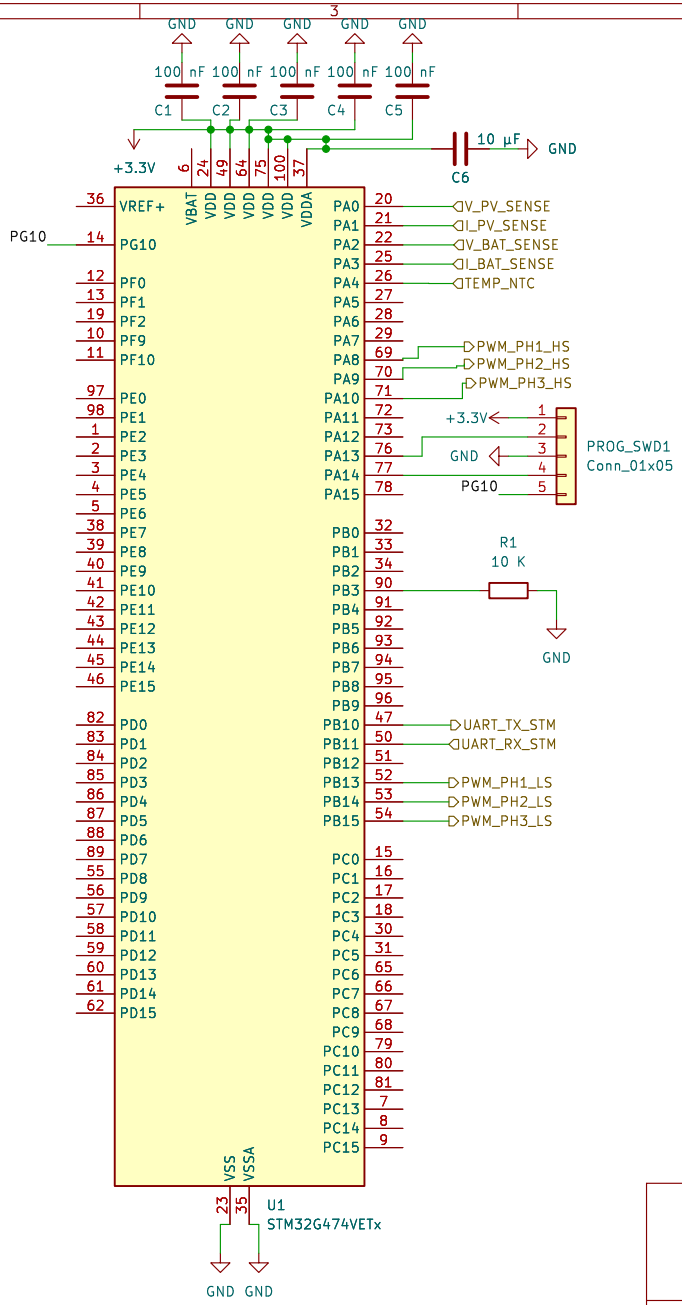
--- 2. RÔLE DES COMPOSANTS PHYSIQUES ---
STM32G474 (MCU)  : "Le Cerveau". Calcule le MPPT et génère les signaux PWM.
UCC21520 (Driver): "Le Muscle". Isole le MCU et amplifie le PWM en 12V.
MOSFET (NMOS)    : "L'Interrupteur". Hache la puissance à 50 000 fois/sec.
ACS758 (Hall)    : "Les Yeux". Mesure les 180A sans contact (magnétique).
Inductance (L)   : "Le Volant d'inertie". Lisse le courant dans chaque phase.
C_ELEC (Low-ESR) : "Le Gros Réservoir". Lisse la tension pour la batterie.
C_CERA (X7R)     : "L'Éponge Rapide". Absorbe les parasites haute fréquence.
```







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  AFFECTATION DES BROCHES – STM32G4
========================================
--- MESURES (ADC) ---
PA0  : V_PV_SENSE  (Tension Solaire)
PA1  : I_PV_SENSE  (Courant Solaire)
PA2  : V_BAT_SENSE (Tension Batterie)
PA3  : I_BAT_SENSE (Courant Batterie)
PA4  : TEMP_NTC    (Sonde Température)

--- COMMANDE PUISSANCE (PWM TIM1) ---
PA8  : PWM_PH1_HS  (Phase 1 – Haut)
PB13 : PWM_PH1_LS  (Phase 1 – Bas)
PA9  : PWM_PH2_HS  (Phase 2 – Haut)
PB14 : PWM_PH2_LS  (Phase 2 – Bas)
PA10 : PWM_PH3_HS  (Phase 3 – Haut)
PB15 : PWM_PH3_LS  (Phase 3 – Bas)

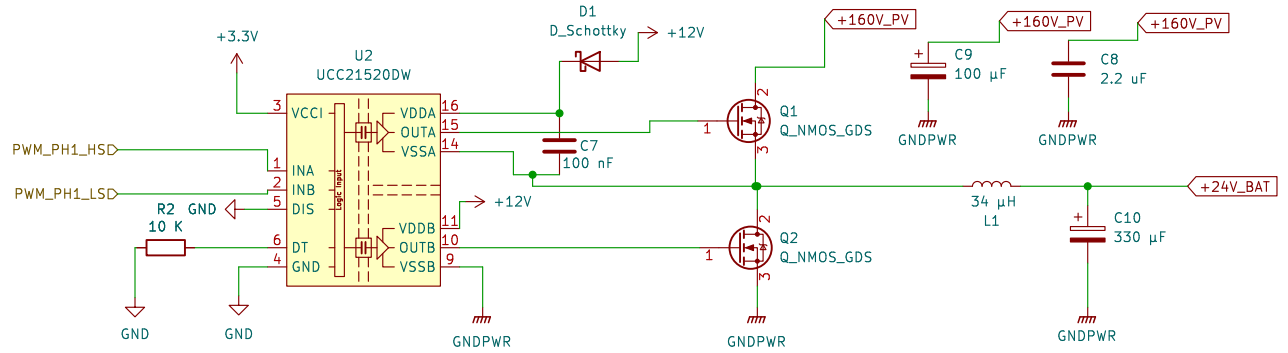
--- PROGRAMMATION & SYSTÈME ---
PA13 : SWDIO       (Data SWD)
PA14 : SWCLK       (Horloge SWD)
PG10 : NRST        (Reset)
P94  : BOOT0       (10k vers GND)

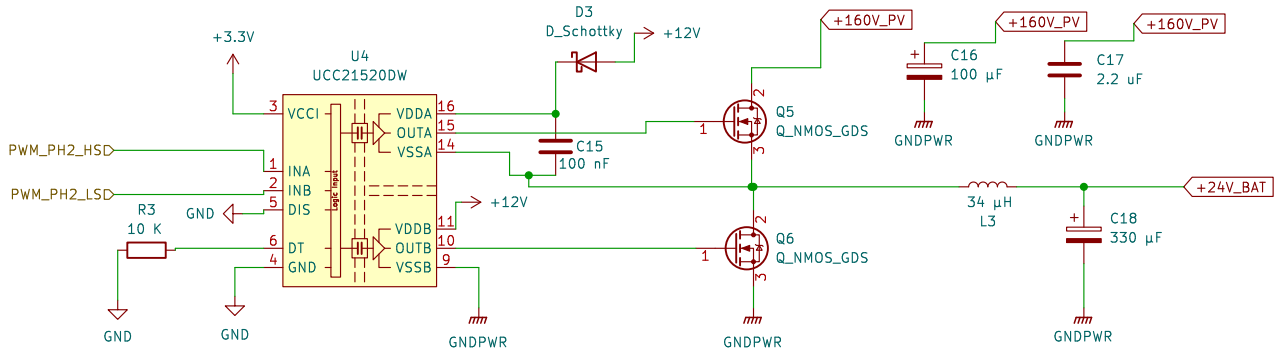
--- Communication UART ---
port USART3
PB10 : TX
PB11 : RX
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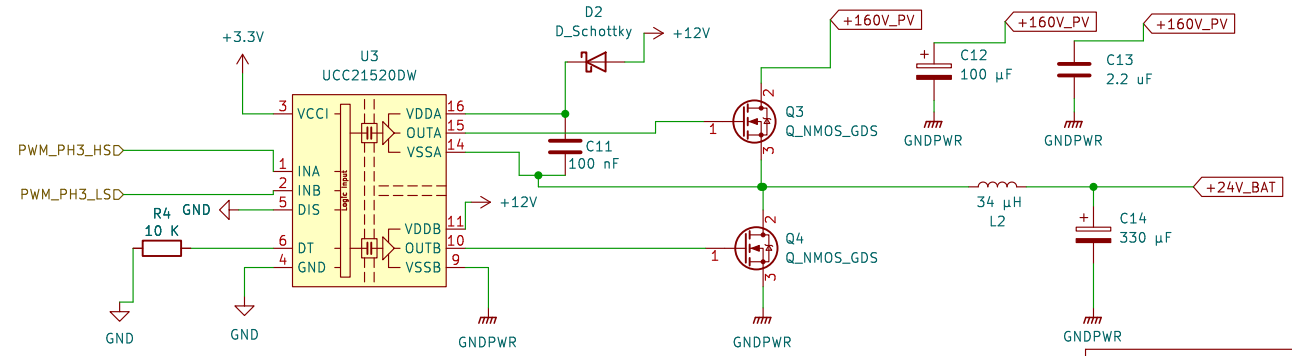
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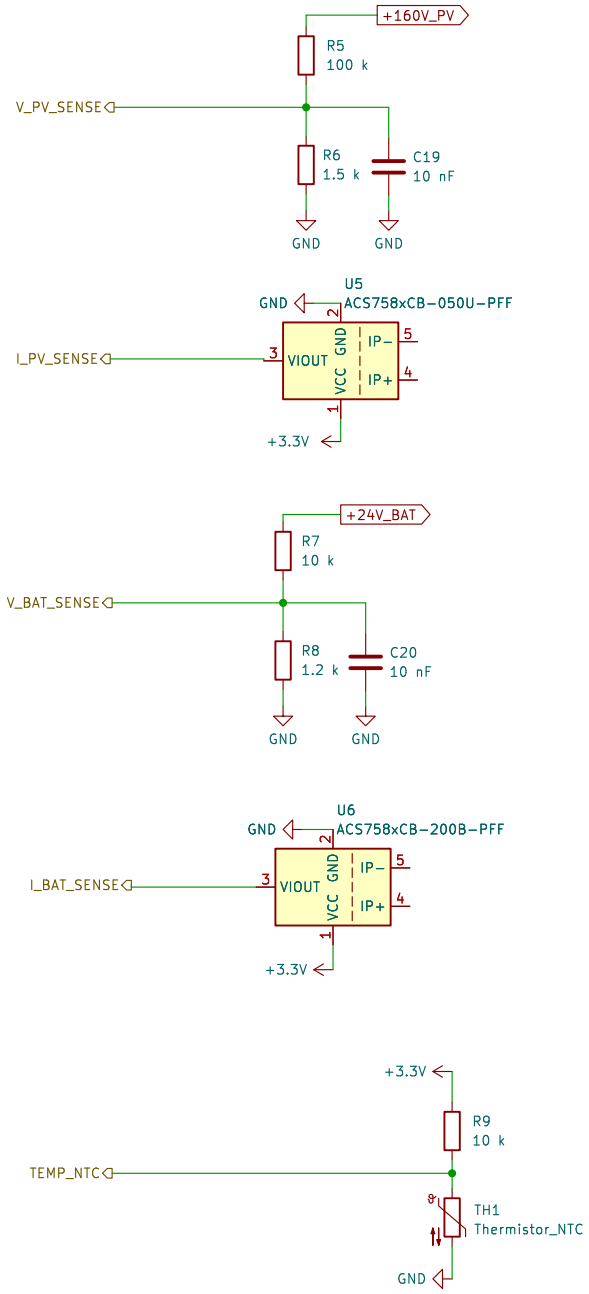




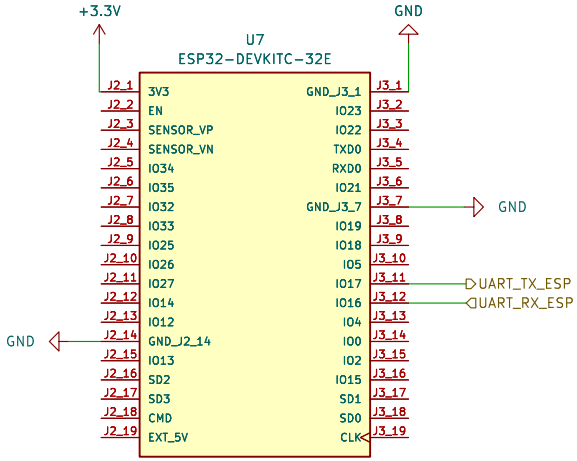
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